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Identifying contemporary challenges pertaining to Distributed Software Development using Fuzzy Delphi Method

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Abstract. Distributed Software Development (DSD) is a process in which team members are located in different geographical locations and collaborate using communication and collaboration tools. DSD has gained prominence in recent years, driven by the surge in globalization, the rapid expansion of the IT outsourcing market, and the widespread accessibility of high-speed internet. However, DSD also presents numerous challenges, including team dynamics, employee well-being, HR management, communication & collaboration, requirements management, tooling & infrastructure, and development process & risk management issues. This paper aims at identifying critical challenges related to DSD. Initially, a systematic literature review was conducted to construct an initial list of 77 challenges; subsequently, a Fuzzy Delphi Method (FDM) survey was adopted to identify the critical challenges in this context. Results of the FDM indicate that, out of the 41 screened-out challenges, differences in work ethics & practices, language barriers, meeting scheduling difficulties, different time zones, burnout, physical fatigue/mental stress, and lack of opportunities for personal development are the most critical challenges of DSD. The findings of this study have implications for both practitioners and researchers related to DSD.

Keywords: DSD, Critical challenges index, Fuzzy Delphi Methodology.

1 Introduction

Distributed Software Development (DSD) has become crucial in the software industry, enabling organizations to utilize global resources and overcome geographical limitations. It allows organizations to expand their human resources, leading to cost savings and increased availability of skilled labor and enables the utilization of expertise from teams in different regions, resulting in a more comprehensive development effort [1]. DSD contributes to shorter development cycles, facilitating faster progress and reduced time to market as well as enhances product integration and improves collaboration and coordination among distributed teams [2].

However, adopting a distributed approach in software development presents challenges such as cultural barriers, knowledge transfer difficulties, and increased

communication overhead [3]. These challenges can impact project organization, control, and product quality; hence, organizations must implement processes and tools to address these challenges and ensure effective coordination and communication among distributed teams [1]. Identifying challenges faced by DSD projects allows researchers and practitioners to gain insights and enhance the success rate of DSD initiatives [4] [5]. This research aims to comprehensively identify critical challenges in DSD through a systematic literature review (SLR) and a Fuzzy Delphi Method screening mechanism. Expert opinions were gathered through a questionnaire to facilitate this process.

This study provides a holistic understanding of the challenges distributed teams face in global software engineering by answering the following questions:

1. What are the contemporary challenges in distributed software development?
2. What are the crucial challenges in distributed software development?

The organization of this study is as follows: Section 2 provides a concise overview of the related work. Section 3 elaborates on the detailed research method. Section 4 presents the results and Section 5 concludes this study.

2 Related Work

The heterogeneity encompassing Distributed Software Development (DSD) pose numerous challenges that leads to practical difficulties in considering many evaluation criteria due to the associated computational and human efforts. Many studies have reported relevant challenges in the DSD domain [1], [2], [4], [6]–[10]. However, these studies exhibit significant limitations as they focus on specific DSD phases or a limited number of challenges within the broader DSD field. Mahmood et al. [7] conducted a systematic literature review addressing the management challenges in DSD projects. Babar et al. [8] should have considered incorporating heuristics for planning, organizing, and managing DSD projects, which are crucial aspects deserving attention. Akbar et al. [9] primarily focused on categorizing the identified challenges in DSD, neglecting specific challenges encountered during DSD projects. In another study [10], the online survey to validate the identified challenges was restricted to agile practitioners, thereby introducing a potential sample bias and possibly failing to adequately represent the perspectives of all stakeholders involved in distributed agile development. Researchers have conducted literature reviews targeting these challenges, only focusing on individual processes, stakeholders, or the entire team [1], [2], [6]–[10]. Some studies have identified less than 30 challenges and have overlooked screening mechanisms and failed to consider contemporary critical challenges, such as improving weight assessment in one study and addressing contemporary challenges in another [9], [10].

A significant gap in the existing literature lies in the need for a systematic screening mechanism to filter out relevant challenges effectively. While previous studies reviewed in the literature have addressed a limited number of challenges, our research has identified a more comprehensive range than any previous study. Furthermore, it is worth noting that contemporary and post-COVID challenges have been largely overlooked in existing studies, emphasizing the need for their investigation. Our research

aims to bridge these gaps and provide a more holistic understanding of distributed teams' challenges in global software engineering.

3 Methodology

This research proposes a methodology to identify critical challenges in distributed software development (DSD). The methodology as illustrated in Figure 1 includes conducting a systematic literature review (SLR) to gather relevant studies and extract DSD challenges. A fuzzy Delphi questionnaire is used to evaluate the importance of these challenges by collecting expert judgments. Barela's Consensus Evaluation assesses the consensus on the critical challenges. These challenges undergo a fuzzy Delphi screening process, resulting in a critical challenges index representing the identified challenges.

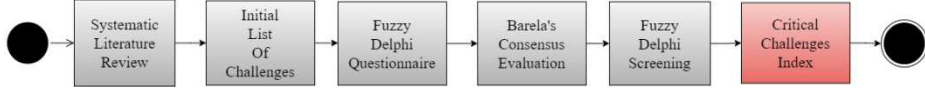


Fig. 1. Methodology

3.1 Preliminary Concepts

Fuzzy Set Theory. Fuzzy set theory, developed by Zadeh et al. [11], extends the classical set theory to handle uncertainties and vagueness in real-world problems. It is particularly useful in multi-criteria decision-making, allowing for meaningful and interpretable representation of ambiguous data. In a fuzzy set, each object is associated with a membership function, denoted as $\mu_{FN}(x)$, which assigns a degree of membership between 0 and 1 to indicate the extent to which the object belongs to the set.

Establishing Fuzzy number. The fundamental definition of fuzzy set theory can be stated as follows: A triangular fuzzy number (TFN), FN is represented by a set (l, m, u) , as illustrated in Figure 2. The membership function $\mu_{FN}(x)$ of FN is defined by equation (1).

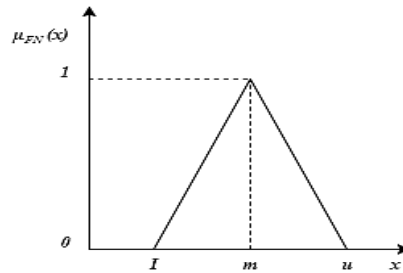


Fig. 2. Triangular Fuzzy Number.

$$\mu_{FN}(x) = \begin{cases} (x-l)/(m-l), & l \leq x \leq m \\ (x-l)/(m-l), & m \leq x \leq u \\ 0, & \text{Otherwise} \end{cases} \quad (1)$$

In the given context, the variables l , m , and u represent crisp numbers that indicate the lowest, most promising, and highest possible values, respectively. These values provide specific points of reference for the TFN [12].

Fuzzy Delphi Method. The Fuzzy Delphi Method (FDM) is valuable for shortlisting critical elements. In this research, Ishikawa et al. [13] proposed FDM that processes expert responses. It offers advantages, including reasonable and objective results with fewer samples and allows adequate expression of expert opinions without distortion [13], [14]. The fuzzy theory addresses uncertainty from differing expert understanding [13]. The geometric mean model for the Fuzzy Delphi method establishes a common understanding among experts [15]. Figure 3 illustrates the overall framework.

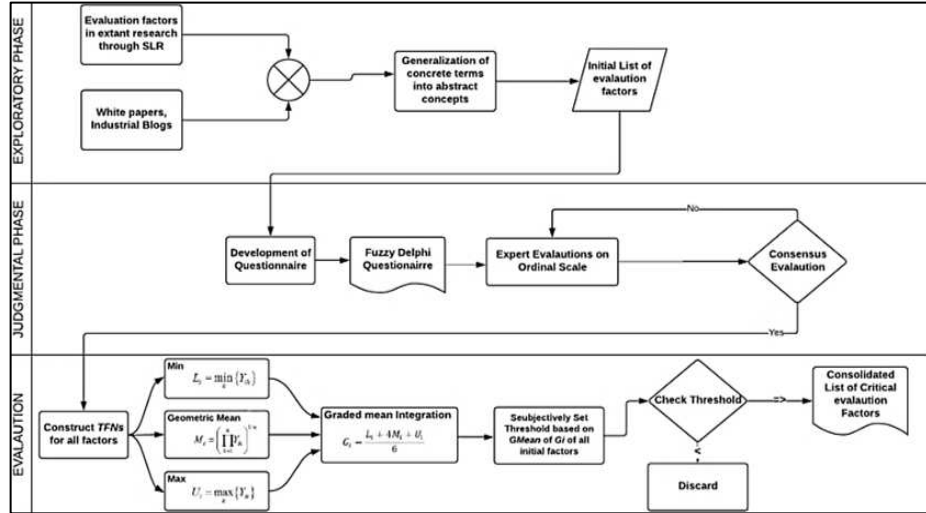


Fig. 3. Fuzzy Delphi Process Flow

Converting expert judgments into estimates. In the first step, expert judgments collected from questionnaires are transformed into estimates, which are then used to generate triangular fuzzy numbers (TFNs).

Here $FN_i (L_i, M_i, U_i)$ can be produced as:

$$L_i = \min_k \{Y_{ik}\} \quad (2)$$

$$M_i = (\prod_{k=1}^n Y_{ik}) \quad (3)$$

$$U_i = \max_k \{Y_{ik}\} \quad (4)$$

Where Y_{ik} refers to the evaluated value of expert opinion k for factor i ; L_i indicates the bottom of the evaluated values. Similarly, M_i and U_i indicate the *geometric mean*, and *ceiling* of the all evaluated values for a factor i .

Screening of critical factors based on threshold value. After obtaining the triangular fuzzy numbers (TFNs), the subsequent step involves screening the factors based on a predefined threshold θ . This screening process is carried out according to specific rules, which are outlined as follows:

If $M_i \geq \theta$, then factor i is Retained.

If $M_i \leq \theta$, then factor i is Dismissed.

3.2 Systematic Literature Review Process

A systematic literature review (SLR) was conducted following Kitchenham et al. guidelines [16] to identify contemporary challenges in distributed software development (DSD). Peer-reviewed articles and relevant grey literature sources were thoroughly examined to understand the topic comprehensively. The SLR process resulted in an initial list of challenges reflecting DSD's diverse and complex nature. This list serves as a foundation for further analysis and exploration, providing insights into the hurdles distributed software development teams face. Building upon the SLR findings, this research aims to address these challenges and propose effective strategies and solutions to enhance the success and efficiency of DSD projects.

3.3 Fuzzy Delphi Questionnaire for Critical Challenges Identification and Barela's Consensus Evaluation

A questionnaire was used to gather expert opinions and assess the importance of challenges in distributed software development (DSD). The questionnaire employed a 1 to 10 ordinal scale for ranking, with 1 as the least important and 10 as the most important. 41 responses were collected from experts. Barela's consensus evaluation method [17] was employed to establish consensus among the experts. This involved analyzing response frequencies and excluding respondents with consistent patterns or extreme viewpoints. After this analysis, 25 responses were retained. Qualified responses were evaluated based on the criterion that expert judgments should fall within two points above and below the mean, ensuring consensus. With a 70% criterion, the consensus evaluation indicated a common consensus among the experts. The 25 responses meeting the consensus criteria underwent the Fuzzy Delphi process, refining and consolidating expert opinions to identify critical challenges in DSD. The Fuzzy Delphi process facilitates aggregating diverse perspectives and generating robust insights into the critical challenges faced in distributed software development.

3.4 Fuzzy Delphi Screening

This phase was conducted using Fuzzy Delphi Method (FDM), the obtained challenges were screened out, explained under the preliminary concepts section.

4 Results and Discussion

This section presents the results and analysis of the study.

4.1 Exploratory Phase

During the exploratory phase of the Fuzzy Delphi Method (FDM), an extensive systematic review of the challenges in Distributed Software Development (DSD) was conducted. Recognizing the practical nature of the problem and the potential divergence between academia and industry perspectives, academic literature and industry sources such as white papers, blogs, and technical forums were analyzed. This approach ensured a comprehensive examination of the challenges in DSD. The convergence of results from both academic and industry sources led to the establishment of an exhaustive list of potential challenges.

However, considering the vast amount of knowledge available, the diversity of DSD, and the influence of multiple dimensions, it is impractical to consider the entire exhaustive list of factors in DSD. Therefore, preliminary generalizations were conducted to identify interrelated and affiliated concepts among the challenges. The framework of the 3 Ps of Product Development was utilized to generalize these challenges. This process facilitated the shortlisting and identification of the most critical evaluation factors in DSD. Figure 4,5 and 6 presents the challenges that resulted from this generalization process.

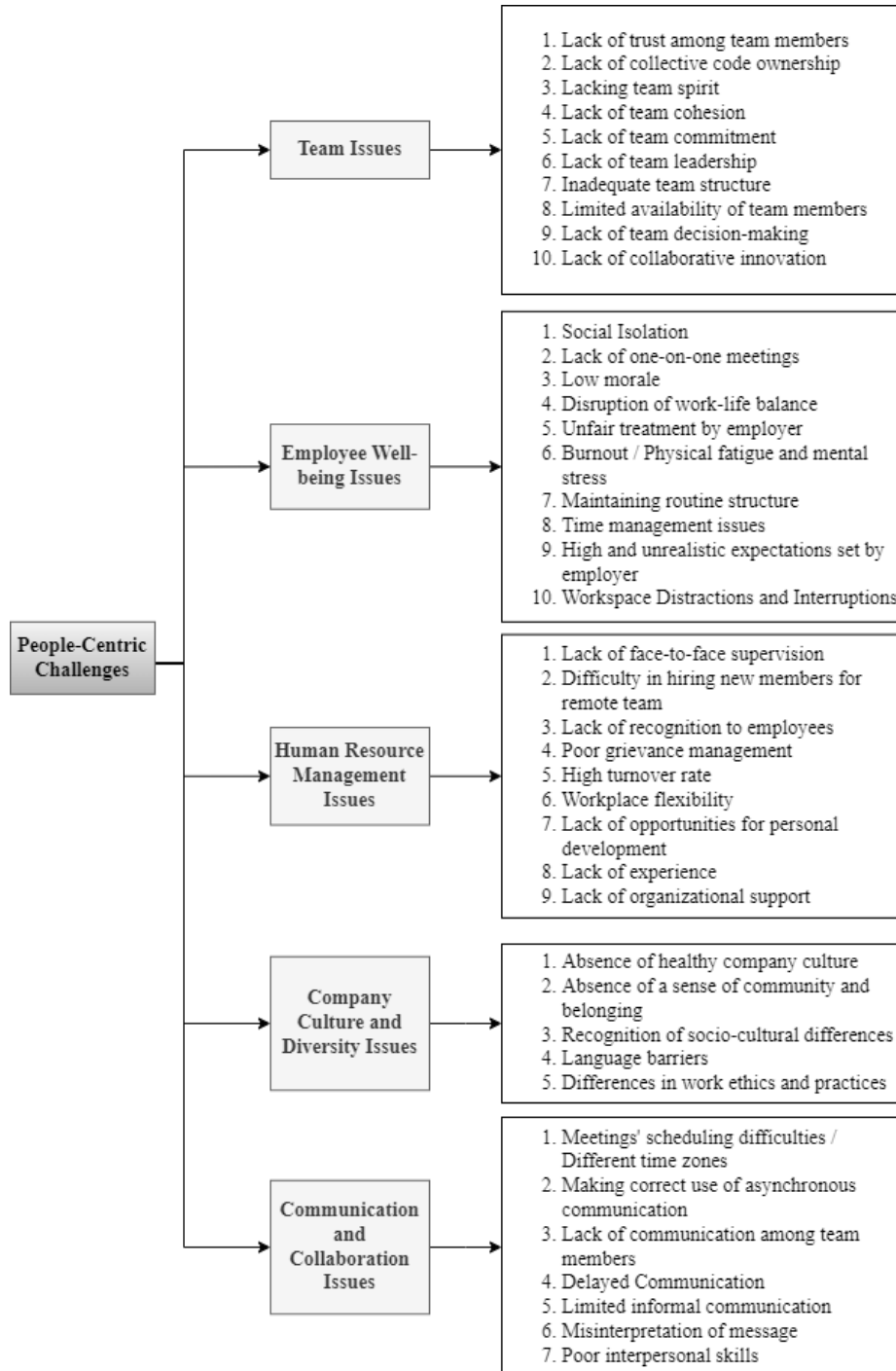
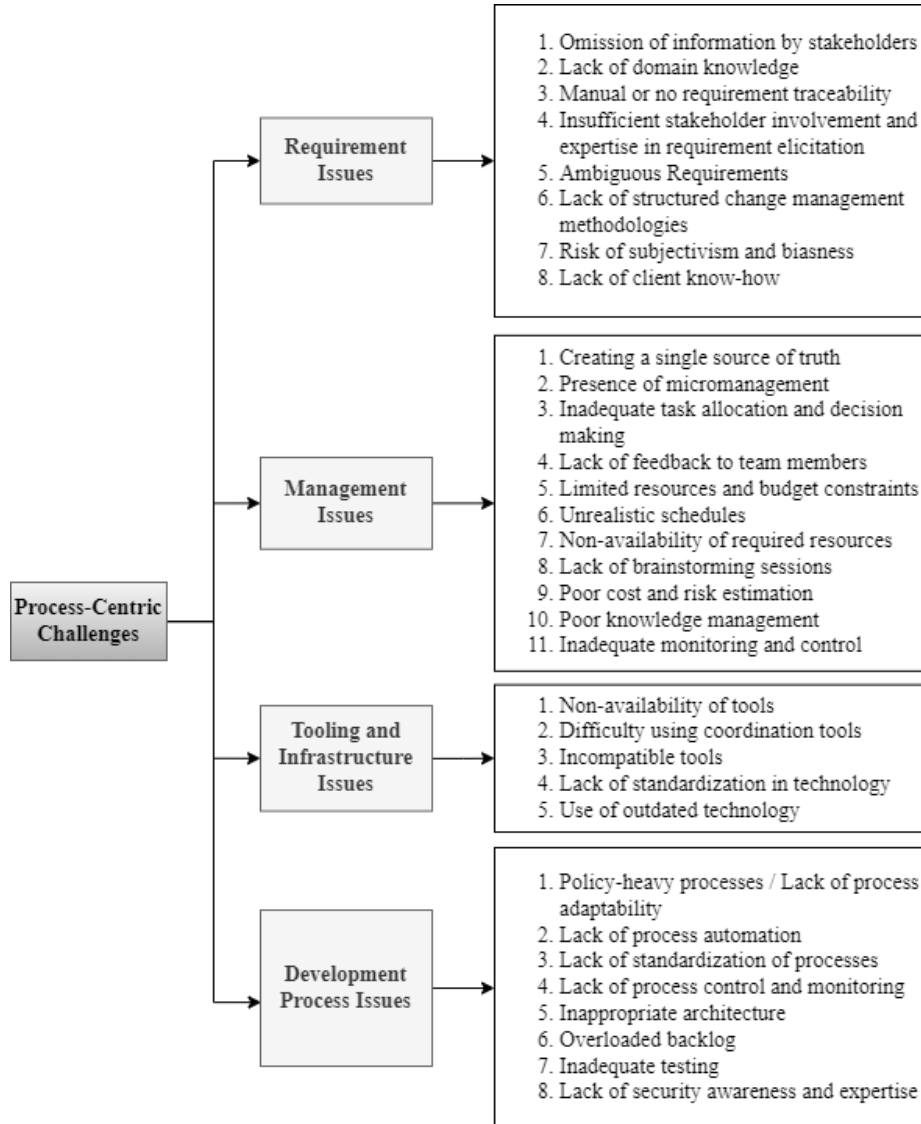


Fig. 4. People-Centric Challenges**Fig. 5.** Process-Centric Challenges

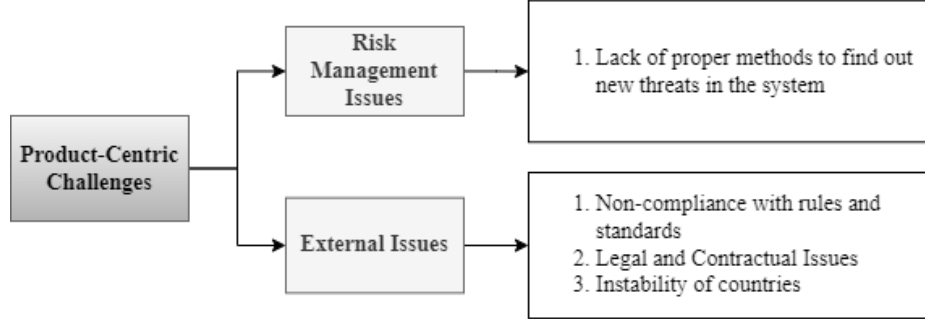


Fig. 6. Product-Centric Challenges

4.2 Judgmental Phase

A questionnaire survey was conducted to gather expert opinions on the significance of evaluation factors related to the identified challenges in DSD. The questionnaire had two sections: one for collecting essential information and demographics and the other for assessing the significance of specific challenges. The experts used an ordinal scale of 1 to 10 to rate the challenges. Out of the 41 received responses, 25 were selected after analyzing frequencies and excluding respondents with consistent patterns or extreme viewpoints. A consensus evaluation was performed using Christie and Barela's method to assess the consensus of experts on each challenge [17].

4.3 Evaluation Phase

The evaluation phase consists of three steps: converting expert judgments into estimates, defuzzification using the Geometric Mean Index Ranking (GMIR) method, and screening critical factors based on a subjective threshold value. This research determined the threshold value subjectively by considering the geometric mean of all evaluation factors, resulting in a threshold value of 5.9. This threshold value is a benchmark for identifying the critical factors in the subsequent analysis. Figure 7 presents the evaluation results for all 77 factors, indicating their relative importance and criticality in distributed software development (DSD).

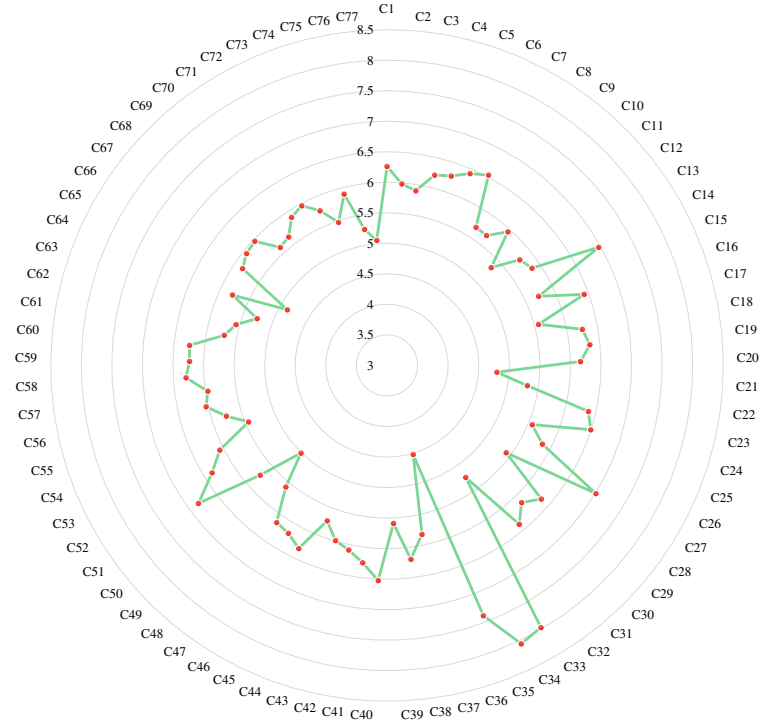


Fig. 7. Initial DSD Challenges with their threshold.

4.4 Synthesis of Results

The results synthesis showcases the rigorous process employed to identify and screen critical challenges in Distributed Software Development (DSD). An extensive research review identified numerous challenges, which were then amalgamated and abstracted into distinct factors. This process resulted in a candidate list of 77 challenge factors. Expert judgments were obtained through a questionnaire survey to assess the significance of these factors. Using a threshold value of 5.9, critical challenges were distinguished from less significant ones based on the geometric mean of all factors. As a result, 41 factors were considered critical challenges, exceeding the threshold, while 36 were dismissed, as presented in Figure 8. The consistency of these findings with established evaluation models and frameworks underscores the research's reliability and validity. This study's index of qualified challenges provides a valuable resource for the DSD research community, enabling more informed decision-making by prioritizing critical project challenges. Organizations can leverage this index to enhance their decision-making processes and improve the success rate of their DSD initiatives.

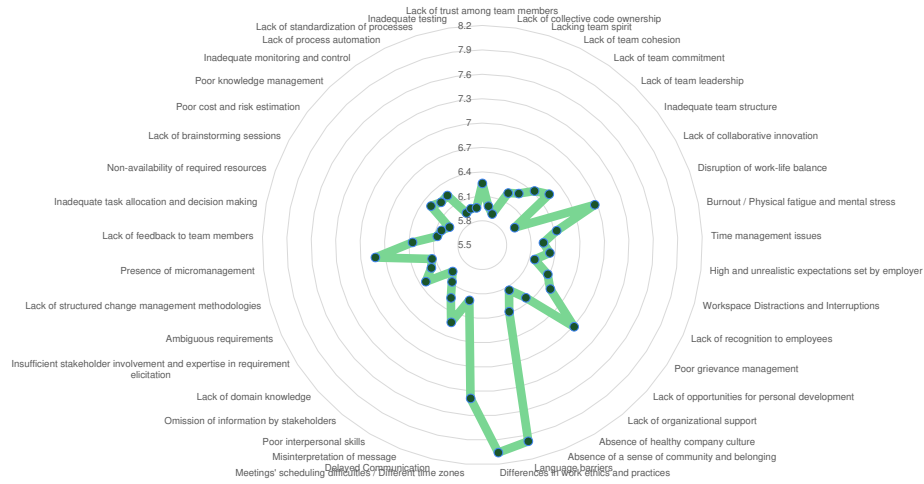


Fig. 8. Qualified Fuzzy Delphi Challenges and their threshold.

Utilizing the Fuzzy Delphi Methodology in this study allowed for a meticulous assessment and categorization of the identified challenges, resulting in three main categories: people, process, and product (3Ps). The comprehensive presentation of these challenges and their corresponding contextual descriptions in Tables 1 and 2 provides valuable insights into DSD's 3Ps aspects. However, the product-centric challenges didn't qualify after FDM. This systematic evaluation serves as a foundational resource for practitioners and decision-makers, offering guidance to develop effective strategies and interventions to address these challenges and elevate the overall DSD process.

Table 1 People-Centric Challenges

S. No.	Challenge	Context	Reference
C1	Lack of trust among team members	When team members do not trust each other, it hinders effective communication, collaboration, and decision-making, decreasing productivity and overall team performance.	[18]–[29]
C2	Lack of collective code ownership	Individual code control in distributed development limits collaboration hampers quality, and slows issue resolution. Shared ownership enhances teamwork, knowledge sharing, and project agility.	[19], [30]

C3	Lacking team spirit	A lack of team spirit within a group can result in reduced motivation, cooperation, and shared goals, impacting the team's ability to achieve objectives and work harmoniously.	[18], [22]
C4	Lack of team cohesion	Insufficient unity and coherence among team members can result in conflicts, misalignment of efforts, and difficulty achieving synergy, hampering the overall team effectiveness.	[10], [18], [21], [22], [26]
C5	Lack of team commitment	When team members are not fully committed to their goals and objectives, it can lead to subpar performance, missed deadlines, and a lack of accountability.	[10], [20]–[22], [26], [27], [31]–[34]
C6	Lack of team leadership	The absence of strong leadership within a team can result in a lack of direction, coordination, and guidance, hindering the team's ability to accomplish tasks efficiently.	[20]–[22], [26], [31]–[33]
C7	Inadequate team structure	Inadequate team structure, such as unclear roles or poor skill alignment, hampers communication and collaboration in distributed software development.	[10], [20]–[22], [26], [31]–[33]
C8	Lack of collaborative innovation	A team needs a collaborative and innovative environment to allow creativity, problem-solving, and new ideas, stifling the team's potential for growth and improvement.	[26], [35], [36]
C9	Disruption of work-life balance	When work demands infringe on personal life, it can result in stress, burnout, and reduced productivity, affecting both the individual team members and the team.	[36]–[38]
C10	Burnout/Physical fatigue and mental stress	Complexity, on-call duties, debugging, and scalability stress engineers in distributed software. Prolonged virtual meetings worsen fatigue, hampering engagement and communication, thus reducing virtual team efficiency.	[33], [37]–[39]
C11	Time management issues	Poor time management within a team can result in missed deadlines, reduced efficiency, and increased stress, affecting the overall team performance.	[26], [37], [38]
C12	High and unrealistic expectations set by employer	Unrealistic expectations can lead to undue pressure on team members, causing stress, low morale, and potential burnout, hindering team productivity and performance.	[31], [32], [37], [40]

C13	Workspace Distractions and Interruptions	Constant disruptions and distractions in the workspace can impede concentration, focus, and productivity, negatively affecting the team's ability to deliver results.	[39]–[41]
C14	Lack of recognition to employees	Failing to recognize and appreciate the efforts of team members can lead to decreased motivation, reduced job satisfaction, and increased turnover rates.	[26], [29], [31], [37], [39]
C15	Poor grievance management	Inadequate handling of employee grievances can lead to dissatisfaction, conflicts, and a hostile work environment, impacting team morale and productivity.	[23], [26], [37], [38], [42], [43]
C16	Lack of opportunities for personal development	When employees lack opportunities for growth and development, it can lead to disengagement, reduced commitment, and hinder overall team performance.	[9], [43]
C17	Lack of organizational support	Insufficient support from the organization can result in decreased resources, limited access to information, and a lack of necessary tools, impacting the team's ability to perform effectively.	[18], [26], [43]
C18	Absence of healthy company culture	A healthy company culture must improve communication, trust, collaboration, and accountability in distributed software development. This undermines productivity, quality, and team cohesion.	[31]
C19	Absence of a sense of community and belonging	In distributed software development, lacking a sense of community and belonging can lead to isolation, reduced collaboration, and lower morale among remote team members.	[21]
C20	Language barriers	Language barriers in distributed software development impede effective communication, causing misunderstandings, delays, and hampering collaboration among remote team members.	[9], [10], [18], [19], [23], [26], [37], [44]–[50]
C21	Differences in work ethics and practices	Differences in work ethics and practices among distributed team members can result in conflicting approaches, hindered coordination, and reduced efficiency in software development.	[9], [18], [37]
C22	Meetings' scheduling difficulties / Different time zones	Scheduling meetings across different time zones in distributed software development can be challenging, leading to coordination issues, delays, and limited real-time collaboration opportunities.	[9], [18], [24], [25], [34], [45]–[47], [49]–[52]

C23	Delayed Communication	Transmission delays can lead to misunderstandings, missed opportunities, and reduced efficiency, affecting the overall collaboration and productivity of the team.	[9], [10], [18], [20], [22]–[29], [31]–[35], [38], [39], [41]–[43], [45]–[73]
C24	Misinterpretation of message	Misinterpretation of communication can lead to conflicts, errors and hinder effective decision-making, impacting the overall team performance.	[9], [10], [18], [20], [22]–[29], [31]–[35], [38], [39], [41]–[43], [45]–[73]
C25	Poor interpersonal skills	Poor interpersonal skills in distributed software development hinder effective communication, collaboration, and relationship-building among remote team members, impacting project outcomes.	[9], [10], [26], [28], [29], [34], [40], [46], [49]

Table 2 Process-Centric Challenges

S. No.	Challenge	Context	Reference
C26	Omission of information by stakeholders	Stakeholders need to provide essential information necessary for effective distributed software development, leading to communication gaps and potential project setbacks.	[18], [19], [23], [30], [51], [59]
C27	Lack of domain knowledge	Team members' insufficient understanding of the project domain hinders decision-making, potentially leading to suboptimal solutions and delays in the development process.	[18], [35], [38], [74]
C28	Insufficient stakeholder involvement and expertise in requirement elicitation	Insufficient stakeholder involvement and expertise in requirement elicitation can lead to incomplete or inaccurate project specifications in distributed software development, resulting in misaligned deliverables and potential rework.	[18], [20], [23], [24], [29], [31], [34], [38], [39], [49], [67]
C29	Ambiguous requirements	Unclear or ambiguous conditions pose significant challenges in distributed software development, leading to misunderstandings, rework, and project inefficiencies.	[18], [27], [34], [47]

C30	Lack of structured change management methodologies	The absence of well-defined change management methodologies can result in difficulties in handling modifications, leading to scope creep and project instability.	[9], [18], [26], [27], [34], [66]
C31	Presence of micromanagement	Excessive micromanagement can impede team autonomy and collaboration, hindering the efficient progress of distributed software development projects.	[43]
C32	Lack of feedback to team members	Inadequate feedback and communication channels within the team.	[18], [23], [27], [29], [38], [41]
C33	Inadequate task allocation and decision making	Inadequate task allocation and decision-making in distributed software development can lead to more explicit responsibilities, project bottlenecks, and delays, affecting overall progress and outcomes.	[26], [38], [41], [61]
C34	Non-availability of required resources	Insufficient access to necessary resources, such as hardware, software, or expertise, can severely impact the team's ability to meet project requirements and deadlines.	[10], [27], [34], [48]
C35	Lack of brainstorming sessions	The absence of brainstorming sessions in distributed software development results in missed creative opportunities, limited idea sharing, and reduced innovation among remote team members.	[26], [35], [36]
C36	Poor cost and risk estimation	Inaccurate cost and risk estimations can lead to budget overruns and unforeseen challenges during the distributed software development lifecycle.	[9], [18], [25], [34], [42], [45], [65], [68], [71], [75]
C37	Poor knowledge management	Inadequate organization and knowledge sharing within the distributed team can result in repeated errors, lost opportunities, and decreased productivity.	[9], [18], [23], [30], [35], [37], [39], [71], [75]
C38	Inadequate monitoring and control	The need for robust monitoring and control mechanisms makes it challenging to track progress, identify issues, and ensure project adherence to timelines and quality standards.	[9], [18], [23], [27], [49], [75]
C39	Lack of process automation	Insufficient automation of repetitive tasks can lead to manual errors, increased development time, and decreased productivity in distributed software development.	[26], [76]

C40	Lack of standardization of processes	The lack of standardization of processes in distributed software development leads to inconsistencies, confusion, and inefficiencies in workflows, impeding collaboration and hindering project advancement.	[10], [49], [76]
C41	Inadequate testing	Inadequate testing in distributed software development can result in undetected bugs, reduced software quality, and compatibility issues due to limited collaboration among remote testing teams.	[26], [27], [34], [35], [42], [47], [61], [71]

5 Conclusion

In conclusion, this research investigated the challenges in distributed software development (DSD) through a systematic literature review, online questionnaire, and consensus evaluation. The study's findings were analyzed and categorized, developing a conceptual taxonomy of the identified challenges. Through the Fuzzy Delphi Method (FDM) screening process, 41 challenges were identified as critical in DSD. These challenges provide valuable insights into crucial areas requiring DSD attention and focus. The developed taxonomy enables industry experts to understand the critical challenges specific to their roles and the overall DSD industry. It is important to note that while this research identifies the critical challenges in DSD, it needs to provide specific mitigation strategies or insights for addressing these challenges. Further research and analysis are necessary to develop practical approaches and strategies for overcoming these challenges and improving the success of DSD projects. Overall, this research contributes to the existing body of knowledge by providing a comprehensive understanding of the challenges in DSD and laying the foundation for future studies and initiatives to address these challenges. It highlights the importance of considering and addressing the critical challenges identified in DSD to enhance the efficiency and effectiveness of software development in distributed environments.

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